

## HM06S006P RoHS Compliant LDMOS RF Power Transistor

### 1. DESCRIPTION

HM06S006P is a MOSFET type transistor specifically designed for VHF/UHF RF amplifier applications.

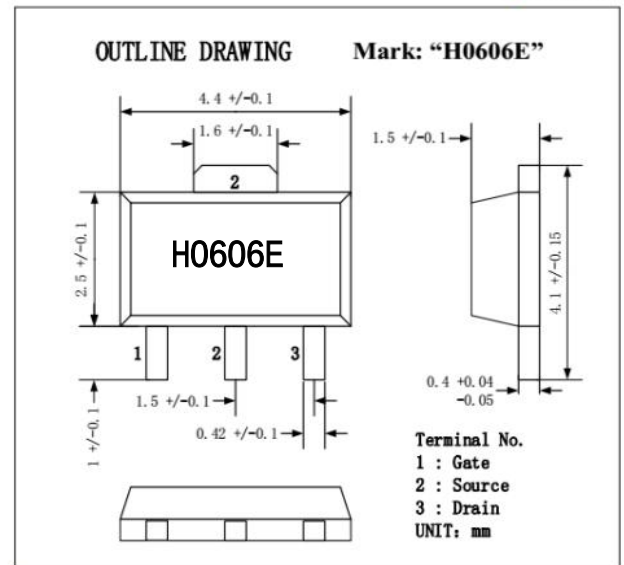
This device has an internal monolithic Zener diode from gate to source for ESD protection.



### 2. FEATURES

- High power:  $P_{out} > 5W$  Typ. @  $V_{DD} = 7.4V$
- High Efficiency:  $\eta_d = 66\%$  Typ. @  $f = 175MHz$
- Integrated gate protection diode

### 3. APPLICATION



- For output stage of high power amplifiers in VHF/UHF Band mobile radio sets.
- For drive stage of high power amplifiers in Universal Broadband.

### 4. ABSOLUTE MAXIMUM RATINGS

(  $T_c = 25^\circ C$  UNLESS OTHERWISE NOTED )

| PARAMETER                      | SYMBOL        | CONDITIONS             | RATINGS   | UNIT         |
|--------------------------------|---------------|------------------------|-----------|--------------|
| Drain to source voltage        | $V_{DSS}$     | $V_{gs} = 0V$          | 20        | V            |
| Gate to source voltage         | $V_{GSS}$     | $V_{ds} = 0V$          | -5 ~ 10   | V            |
| Operating Voltage              | $V_{DD}$      | -                      | 9         | V            |
| Drain Current                  | $I_D$         | -                      | 3.2       | A            |
| Input Power                    | $P_{in}$      | $Z_g = Z_l = 50\Omega$ | 1500      | mW           |
| Storage temperature            | $T_{stg}$     | -                      | -55 ~ 150 | $^\circ C$   |
| Operating Junction Temperature | $T_J$         | -                      | -40 ~ 150 | $^\circ C$   |
| Thermal resistance             | $Z_{th(j-c)}$ | Junction to case       | 6.5       | $^\circ C/W$ |

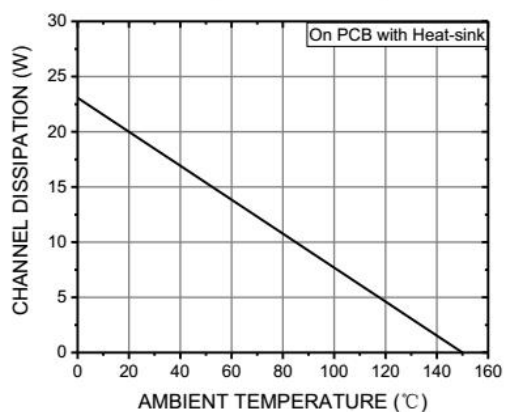
## 5. ELECTRICAL CHARACTERISTICS

(Tc=25°C UNLESS OTHERWISE NOTED)

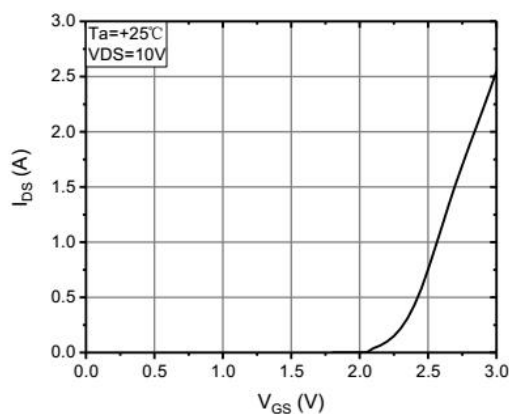
| PARAMETER                               | SYMBOL        | CONDITIONS                                                                                    | MIN                   | TYP  | MAX | UNIT    |
|-----------------------------------------|---------------|-----------------------------------------------------------------------------------------------|-----------------------|------|-----|---------|
| Breakdown Voltage                       | $V_{(BR)DDS}$ | $V_{GG} = 0V, I_D = 39.6\mu A$                                                                | 20                    | -    | -   | V       |
| Zero Gate Voltage Drain Leakage Current | $I_{DSS}$     | $V_{DD} = 12V, V_{GG} = 0V$                                                                   | 1                     | -    | -   | $\mu A$ |
| Gate-Source Leakage Current             | $I_{GSS}$     | $V_{DD} = 0V, V_{GG} = 10V$                                                                   | 1                     | -    | -   | $\mu A$ |
| Gate Threshold Voltage                  | $V_{th}$      | $V_{DD} = V_{gg}, I_d = 39.6\mu A$                                                            | 0.8                   | 1.55 | 2.6 | V       |
| Output Power                            | $P_{out}$     | $V_{DD} = 7.4V, P_{in} = 0.31W$<br>$f = 175MHz, I_{dq} = 500mA$                               |                       | 6.5  |     | W       |
| Drain Efficiency                        | $\eta_d$      |                                                                                               |                       | 66   |     | %       |
| Output Power                            | $P_{out}$     | $V_{DD} = 7.4V, P_{in} = 0.31W$<br>$f = 520MHz, I_{dq} = 500mA$                               |                       | 5.5  |     | W       |
| Drain Efficiency                        | $\eta_d$      |                                                                                               |                       | 61   |     | %       |
| Load Mismatch                           |               | $V_{DD} = 7.4V, P_o = 5W$<br>$f = 175MHz, I_{dq} = 500mA$<br>Load VSWR = 10:1<br>(All phase)  | No Device Degradation |      |     |         |
| Load Mismatch                           |               | $V_{DD} = 7.4V, P_o = 5W$<br>$F = 520MHz, I_{dq} = 500mA$<br>Load VSWR = 10:1<br>(All phases) | No Device Degradation |      |     |         |

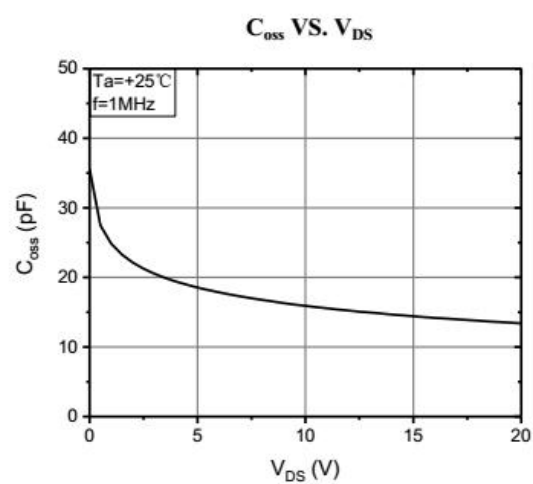
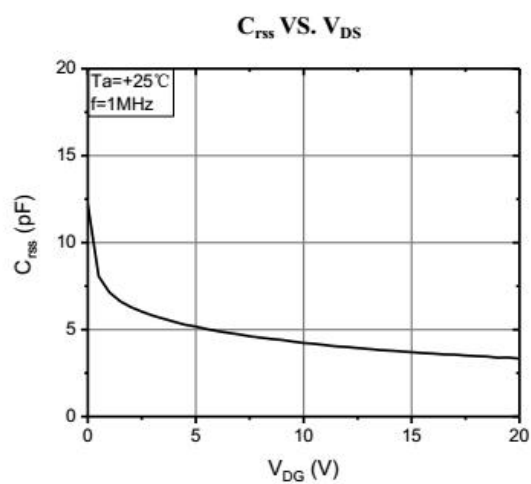
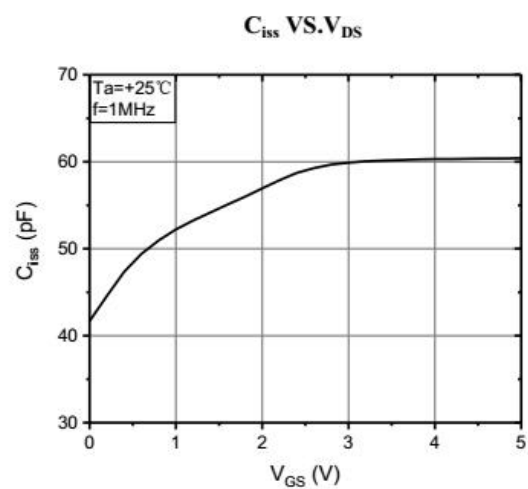
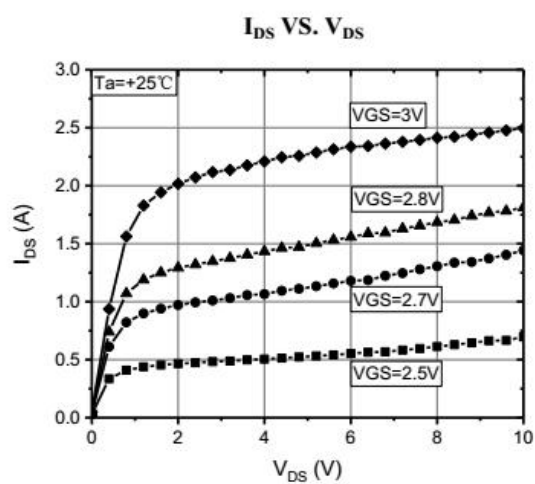
## 6. TYPICAL CHARACTERISTICS

CHANNEL DISSIPATION VS.  
AMBIENT TEMPERATURE

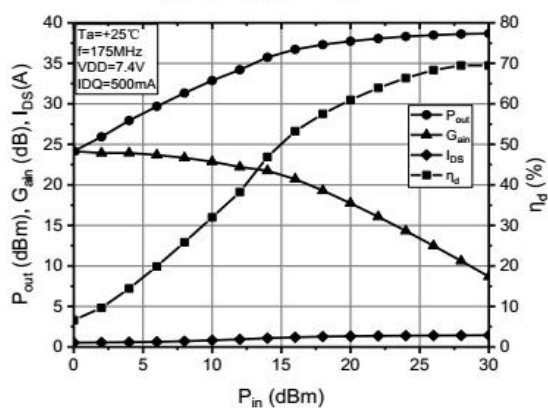


$I_{DS}$  VS.  $V_{GS}$

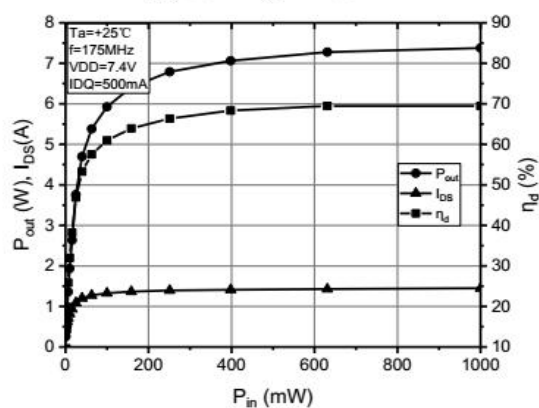




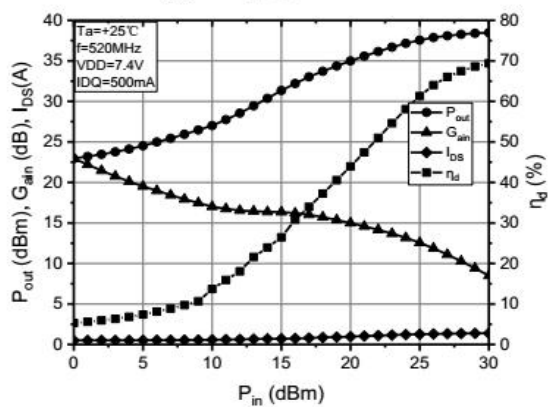
$P_{out}$  VS.  $P_{in}$  @  $f = 175\text{MHz}$



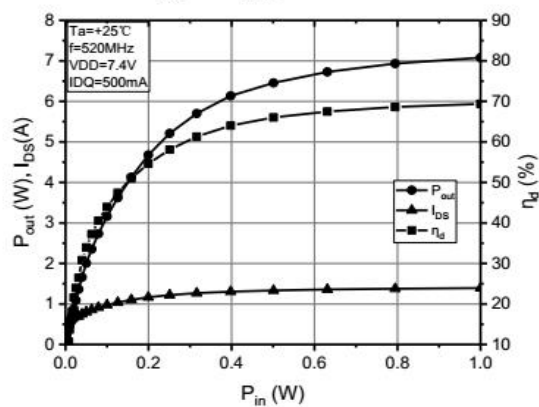
$P_{out}$  VS.  $P_{in}$  @  $f = 175\text{MHz}$



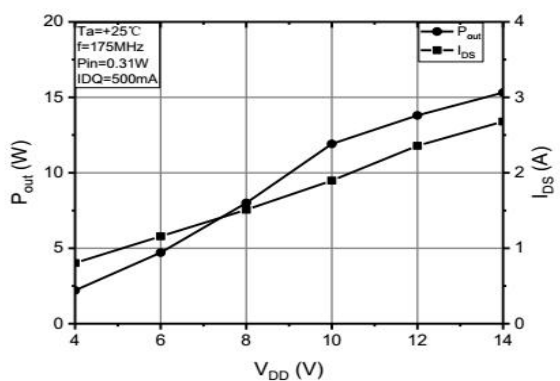
$P_{out}$  VS.  $P_{in}$  @  $f = 520\text{MHz}$



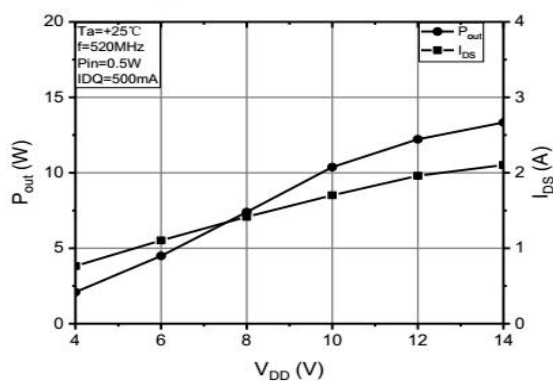
$P_{out}$  VS.  $P_{in}$  @  $f = 520\text{MHz}$

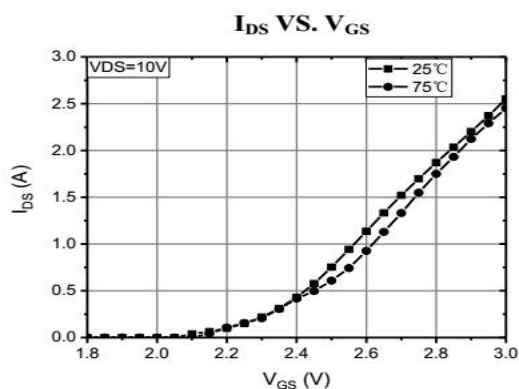


$P_{out}$  VS.  $V_{DD}$  @  $f = 175\text{MHz}$



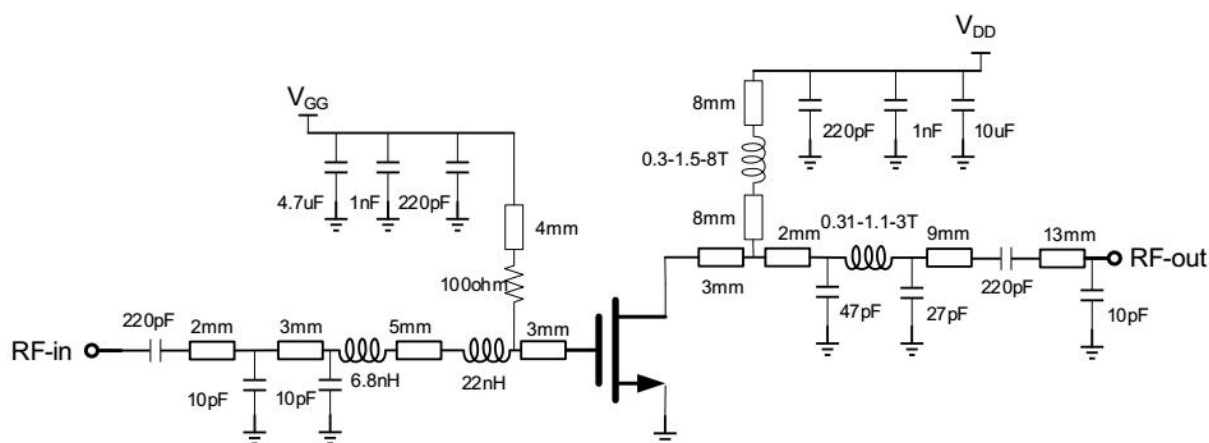
$P_{out}$  VS.  $V_{DD}$  @  $f = 520\text{MHz}$





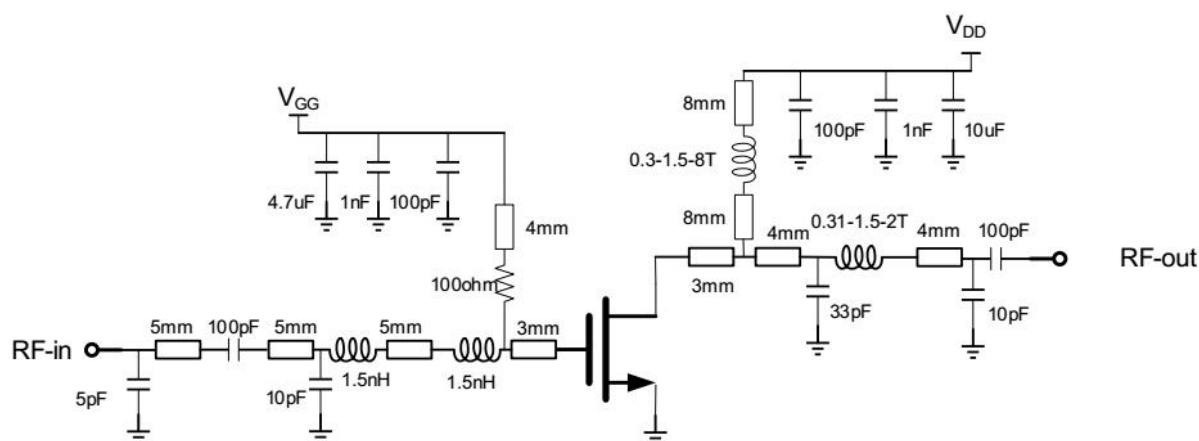
## 7. TEST CIRCUIT

**175MHz @  $V_{DD} = 7.4V$ ,  $I_{dq} = 500mA$**



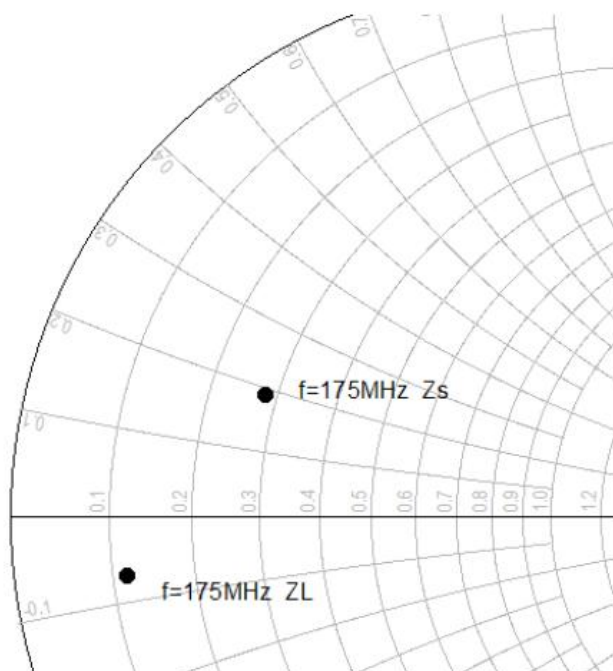
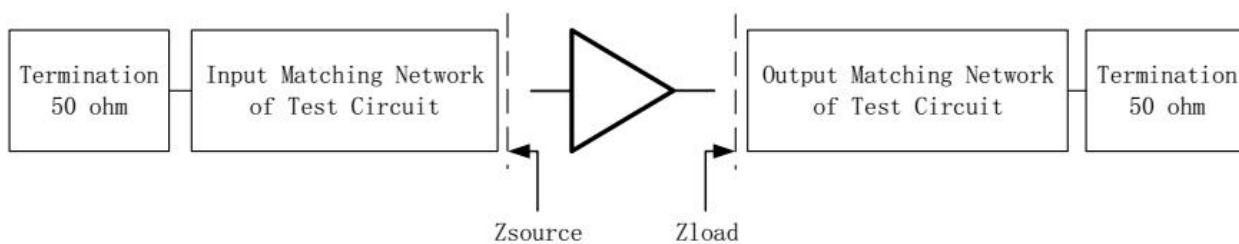
Note: The characteristic impedance of all microstrip lines: 50ohm

**520MHz @  $V_{DD} = 7.4V$ ,  $I_{dq} = 500mA$**

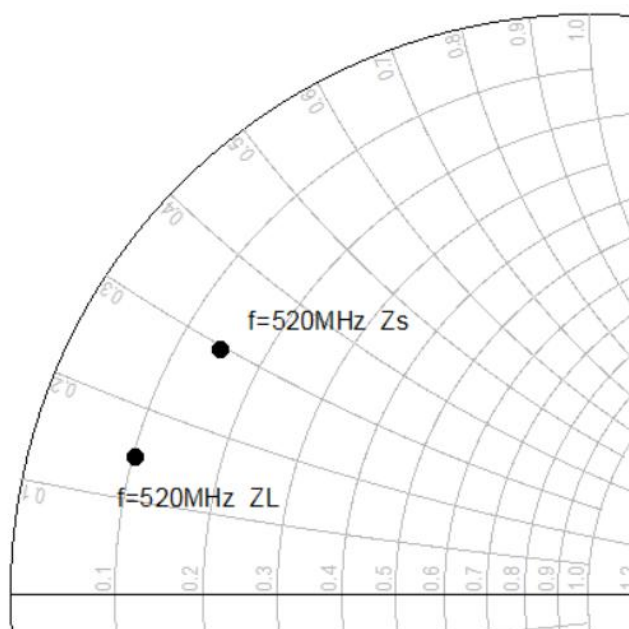


Note: The characteristic impedance of all microstrip lines: 50ohm

## 8. INPUT/OUTPUT IMPEDANCE VS. FREQ. CHARACTERISTICS



| @ $P_{in}=0.31\text{W}$ , $V_{DS}=7.4\text{V}$ , $I_{dq}=500\text{mA}$ |                  |
|------------------------------------------------------------------------|------------------|
| $Z_s$                                                                  | $13.90 + j 9.38$ |
| $Z_L$                                                                  | $5.21 - j 3.41$  |



| @ $P_{in}=0.5\text{W}$ , $V_{DS}=7.4\text{V}$ , $I_{dq}=500\text{mA}$ |                  |
|-----------------------------------------------------------------------|------------------|
| $Z_s$                                                                 | $7.02 + j 15.30$ |
| $Z_L$                                                                 | $4.92 + j 7.42$  |

9、HM06S006P S-PARAMETER DATA ( $V_{DD}=7.4V$ ,  $I_{dq}=500mA$ )

| Freq<br>(MHz) | S11   |       | S21   |        | S12   |        | S22   |        |
|---------------|-------|-------|-------|--------|-------|--------|-------|--------|
|               | (mag) | (ang) | (mag) | (ang)  | (mag) | (ang)  | (mag) | (ang)  |
| 150           | 0.99  | 176.0 | 7.56  | 3.5    | 0.02  | -83.3  | 0.84  | 15.1   |
| 175           | 0.97  | 172.4 | 6.66  | -11.2  | 0.02  | -97.2  | 0.84  | -10.1  |
| 200           | 0.96  | 167.8 | 5.62  | -32.9  | 0.02  | -117.5 | 0.84  | -46.2  |
| 250           | 0.94  | 161.2 | 4.44  | -68.0  | 0.02  | -151.7 | 0.84  | -106.0 |
| 300           | 0.93  | 155.3 | 3.66  | -102.2 | 0.02  | 175.7  | 0.84  | -165.2 |
| 350           | 0.92  | 149.6 | 3.11  | -135.8 | 0.02  | 144.3  | 0.83  | 136.3  |
| 400           | 0.92  | 144.2 | 2.70  | -169.0 | 0.02  | 112.3  | 0.83  | 78.0   |
| 450           | 0.92  | 138.9 | 2.39  | 157.8  | 0.01  | 80.5   | 0.83  | 19.7   |
| 500           | 0.91  | 133.7 | 2.16  | 124.6  | 0.02  | 50.4   | 0.83  | -38.6  |
| 520           | 0.91  | 131.6 | 2.07  | 111.2  | 0.02  | 37.5   | 0.84  | -62.2  |
| 550           | 0.91  | 128.6 | 1.95  | 91.2   | 0.02  | 18.2   | 0.83  | -97.2  |
| 600           | 0.91  | 123.4 | 1.77  | 58.0   | 0.01  | -12.6  | 0.83  | -155.8 |
| 650           | 0.91  | 118.4 | 1.62  | 25.2   | 0.01  | -43.5  | 0.82  | 146.2  |
| 700           | 0.90  | 113.4 | 1.49  | -7.2   | 0.01  | -73.9  | 0.82  | 88.8   |
| 750           | 0.90  | 108.5 | 1.39  | -39.7  | 0.01  | -102.5 | 0.82  | 31.7   |
| 800           | 0.90  | 103.6 | 1.30  | -72.3  | 0.01  | -134.1 | 0.82  | -25.7  |
| 850           | 0.89  | 98.7  | 1.23  | -105.1 | 0.01  | -164.7 | 0.82  | -83.8  |
| 900           | 0.89  | 93.9  | 1.16  | -138.2 | 0.01  | 162.4  | 0.81  | -142.6 |
| 950           | 0.89  | 89.1  | 1.09  | -171.1 | 0.01  | 131.3  | 0.81  | 158.6  |
| 1000          | 0.88  | 84.5  | 1.02  | 156.6  | 0.01  | 107.3  | 0.80  | 100.7  |
| 1050          | 0.87  | 80.0  | 0.97  | 124.3  | 0.01  | 72.7   | 0.80  | 43.9   |
| 1100          | 0.87  | 75.7  | 0.92  | 92.5   | 0.01  | 42.5   | 0.79  | -12.8  |
| 1150          | 0.87  | 71.5  | 0.89  | 60.4   | 0.01  | 12.7   | 0.80  | -70.4  |
| 1200          | 0.87  | 67.0  | 0.86  | 27.8   | 0.01  | -18.1  | 0.79  | -128.5 |
| 1250          | 0.87  | 62.5  | 0.83  | -4.9   | 0.01  | -49.3  | 0.79  | 172.8  |
| 1300          | 0.87  | 57.9  | 0.80  | -37.5  | 0.01  | -79.2  | 0.78  | 114.6  |
| 1350          | 0.87  | 53.3  | 0.78  | -70.0  | 0.01  | -108.3 | 0.78  | 56.6   |
| 1400          | 0.87  | 48.8  | 0.75  | -102.4 | 0.01  | -140.5 | 0.79  | -0.8   |
| 1450          | 0.86  | 44.3  | 0.74  | -135.0 | 0.01  | -172.2 | 0.79  | -58.6  |
| 1500          | 0.86  | 39.8  | 0.72  | -167.7 | 0.01  | 158.4  | 0.78  | -117.6 |

## 10. RECOMMENDED PCB PAD LAYOUT

